

1. Introduction

The Satellite Learning Laboratory is a satellite engineering ecosystem that supports workforce development, training and hands on experimentation. It also provides a tool to support the development of spacecraft subsystems, hardware, software and operations procedures.

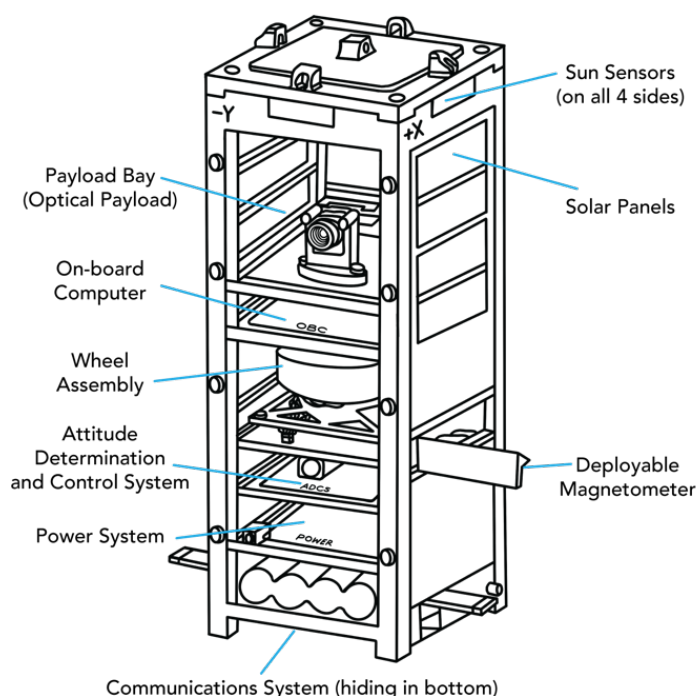
2. Aims and objectives

This lab will familiarise the user with the Satellite Learning Laboratory Ecosystem.

3. Equipment needed

The following equipment will be required for this lab:

- Satellite Simulator
- Remove Before Flight Power Inhibit Key
- Launch Dispenser
- Multimeter and probes
- Power adapter
- Ground station and lead
- PC / laptop with SCOTTI installed



4. Prerequisites

None.

5. Background reading and preparation

If possible, familiarise yourself with the Quick Start Guide or the User Manual for the Satellite Simulator.

6. Familiarisation

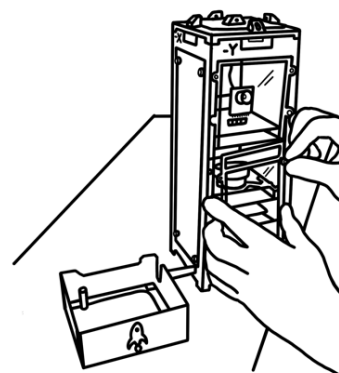
The Satellite Simulator includes functionally representative electronic assemblies for each of the main elements of the satellite:

- Communications System
- Electrical Power System
- Attitude Determination and Control System
- On-Board Data Handling System
- Optical Payload

Most of the satellite platform subsystems are in the bottom part of the Satellite Simulator structure, leaving the top area for accommodation of the payload. The Satellite Learning Laboratory Standard Kit includes imaging payload based on an OV2620 2MP ArduCam sensor.

The Satellite Simulator has been designed to support a range of use cases, some of which require access to the internals of the Satellite Simulator and the ability to remove and fit subsystems and components.

The front of the Satellite Simulator has an access panel for the platform subsystems and payload bay, and the back has a payload bay access panel. There are four thumbscrews on each panel, removal of which enables the access panel to be removed. This will be covered in a later experiment.



Warning - Removal of the access panels exposes the inner workings of the Satellite Simulator and should only be carried out by qualified persons or under the supervision of a qualified person. Observe ESD precautions, beware of moving parts, hot components and the risk of electric shock when removing or fitting components.



6.1 Turning the Satellite Simulator on and off

The Launch Dispenser and Remove Before Flight Power Inhibit Key are used to turn on and off the power to the Satellite Simulator.

Fitting the Remove Before Flight Power Inhibit Key into the Satellite Simulator and placing the Satellite Simulator into the Launch Dispenser will turn the power off. In this state none of the indicators within the Satellite Simulator should be illuminated.

The only exception will be if the unit is being charged and the external power source is connected, in which case the charge indicator may be illuminated.

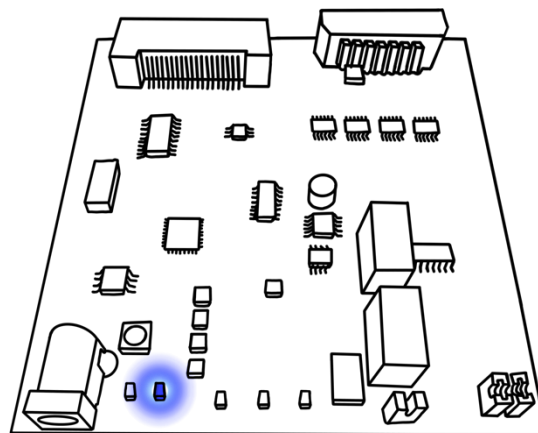
Removal of the Remove Before Flight Power Inhibit Key while the Satellite Simulator is still in the Launch Dispenser does not change the state of the unit, all indicators should be off.



Removing the Remove Before Flight Power Inhibit Key from the Satellite Simulator and then removing the Satellite Simulator from the Launch Dispenser, will place the Satellite Simulator in deployed mode.

A single blue indicator will be illuminated signifying that core elements of the power system have power and that the deployment timer has been initiated.

After a short period, other indicators will be illuminated, indicating that the Satellite Simulator is now in an operational state.



7. Deployment Exercise

The first exercise with the Satellite Simulator is to simulate the launch of a satellite.

Start with the Satellite Simulator with all its remove before flight items inserted, and have the Launch Dispenser ready.

When a satellite arrives at a launch site, it will have remove before flight items attached to protect various equipment and to make it safe for delivery, e.g. power system off.

When the satellite is to be integrated to the launch vehicle there are a sequence of activities to perform such as, filling up fuel tanks, charging the battery, removal of remove before flight equipments.

For this exercise, perform the following steps:

1. Remove the "Remove Before Flight Items" (the power inhibit and the lens cap).
2. Place the Satellite Simulator in the Launch Dispenser.

Now the satellite simulator is ready for "launch". Now perform the following step to launch the Satellite Simulator:

3. Remove the Satellite Simulator from the Launch Dispenser.

Now record the following information:

4. The sequence for the subsystems turning on post deployment
5. The time it takes for the power subsystem to start post deployment
6. Which systems do not automatically turn on

Discuss and record the following:

7. Why might you delay the satellite turning on post deployment?
8. Why might you not want all the equipment to turn on post deployment?

8. Communications Exercise

For the rest of this lab, we need to use SCOTTI to communicate with the Satellite Simulator.



Note

Each Satellite Simulator and ground station radio is preconfigured with a default operating frequency, paired for use. If there are multiple units, check the frequency of the ground station matches that of the Satellite Simulator. The channel number for the Satellite Simulator is on the bottom of the unit.



1. Plug in the ground station.
2. Start the SCOTTI software.
3. Click "Yes" to load the default database
4. Select the "Comm" menu tab.
5. Go to the "Select Port" option
6. Choose the USB port the ground station is plugged into.
7. Select the "Connect" option



You should now be connected to the Satellite Simulator.

You can now explore the different screens and information provided. Record the answers to the following questions:

8. What are the different Roles?
9. What information can you get from the operations page?
 - a. What is the battery voltage?
 - b. Illuminate the solar panels with a torch, what voltage do the solar panels reach?
10. From which screen can you turn the payload and AOCS equipment on?
11. On which screen can you see the gyro telemetry values?
 - a. What are they?
12. Where can you find temperature information?
 - a. What is the temperature of the backplane?
13. Which screen shows the raw data coming off the satellite?
 - a. How many packets have been received at this point?
 - b. How many incomplete packets have been received?
 - c. Which subsystems are sending telemetry?

The table on the next page shows the Roles and associated pages that can be found within SCOTTI.



Role: Operations

Overview of key systems status.
Control of some ADCS and payload

Role: Sub System Tester ADCS Page

ADCS telemetry window
Sensor positions
Mode selection and wheel control
Torque rod control

Command buffer and Send/Clear option

Role: Sub System Tester Power Page

Power system telemetry
System power
Power switches for: platform, ADCS, payload and thermal experiment

Command buffer and Send/Clear option

Role: Sub System Tester Thermal Page

Thermal telemetry
Thermistor values

Command buffer and Send/Clear option

Role: Developer

Raw data stream.
Packet information.



SATELLITE LEARNING LABORATORY

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Version : 1.3, Date : June 2025